DREAM
Danish Research Institute for
Economic Analysis and Modelling

GREEN REFORM

Project activities and timeline for GreenREFORM EU Version 2.0

Project plan version 2

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Economic memo
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Context

This document presents an updated project plan, cf. the following quote from the project proposal submitted to the KR Foundation August 2025:

"At the initial phase of the project, each of the four institutions will be asked to produce an updated plan for data development and implementation, identifying the need for support and enabling a revision of the project plan."

Project objectives

The plans for data development and implementation referred to above has been delivered. In brief, all four institutions remain committed and are requesting support from DREAM as anticipated. Further, the ambition of each institution is well aligned with the indicators of success of the project objective to enhance the capacity for climate policy analysis of the four institutions. A combined summary of data and implementation plans are provided in appendix.

Progress is also being made on the project objective to foster engagement from additional countries and support the design of new, more ambitious projects. Specifically, DREAM has signed a partnership agreement with the Development Economics Research Group at The University of Copenhagen to partner in a project with the objective of adapting and calibrating the GreenREFORM EU model for implementation in Vietnam. We are very pleased, that The GreenREFORM EU 2.0 project allows us to allocate resources towards supporting the much larger efforts of the project team at the University of Copenhagen. The partnership agreement is provided in appendix for further context[1].

[1] In the partnership agreement, DREAM commits to providing finance for one year of a Phd position. This however is outside the scope of the GreenREFORM EU 2.0 project.

It is also worth noting, that the EU TSI 2026 project application mentioned in the project proposal has gained support by institutions in no less than 18 EU member states, and has been pre-qualified for support by the EU Commission. At present, the TSI project is expected to commence in September 2026. In consequence, we aim at completing the current project October to avoid too much of an overlap between the two projects. - As envisioned in the project proposal, the GreenREFORM EU 2.0 will ensure a smooth transition with continuous model development and support for the capacity-building efforts of the participants

Activities

The headline activities of the project remain unchanged, except for the partnership with the University of Copenhagen (UCPH), which we define as a standalone headline activity.

At the level of project management, we have been working towards the preliminary achievements as described above, and obviously towards the present project plan revision.

In the first version revision of the project plan, it was anticipated, that the model documentation would be done at the end of the project. An unexpected availability of internal resources has made it opportune to commence on this already. It will be beneficial to have the documentation finished before the TSI 2026 project commence.

Resources are also being allocated to support especially Belgium but also Italy in calibrating the model to their own data. Preliminary results are promising, but there is still much work to be done.

Headline activities:

- Model development
- Capacity Building.
- Data and implementation
- *Support of UCPH Vietnam project*

Model development:

Under the headline of model development work on the theoretical documentation of the model is in progress, as stated above. Also, a prioritized list of desired model refinements has been drafted in discussion with the four partner institutions. The list is available in appendix. We aim to complete as many of the model refinements as possible within the time and budgetary constraints of the project. Especially, there is a great deal of uncertainty regarding resources needed for direct support, which is a priority.

Capacity Building:

For now, capacity building is not a priority, as the four institutions are all focused on data development. Hence, no specific activities have been carried out or planned at present.

Data and implementation:

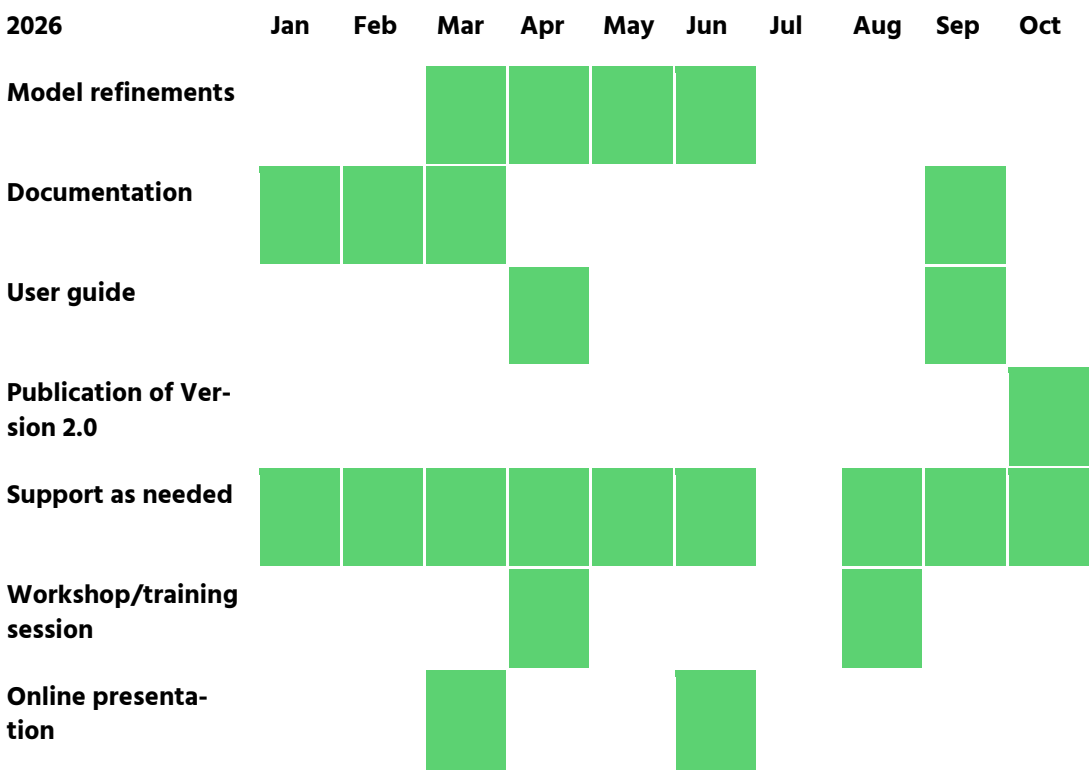
The data and implementation plans received from the four partner institutions help us allocate time and budget, as reflected below.

Support of UCPH Vietnam project

The partnership with UCPH is in its early stages. We don’t have a clear picture of the activities needed to be carried, and the budget allocation is thus also quite preliminary.

Timeline

An updated timetable is provided below. As stated, we aim at completing all activities by October and not end of year as in the project proposal.



Budget allocation

The allocation and resources spent to date on each of the headline activities appears from the first table below.

Half of the budget is allocated towards Model development, which includes the activities Documentation and Model refinements, cf. the timeline table. A little less than a third of the budget is allocated towards Data and implementation, which includes all activities related to the direct support of the four partner institutions.

Headline activity	Allocation, days	Time spend, days
Model development	56.5	4.6
Capacity Building	5	0.0
Data and Implementation	30	4.4
Project admin etc.	5	2.2
Support of UCPH Vietnam project	5	0.9
	101.5	**12.2**

The second table brakes down allocation and time spend on the headline activity Model development on each of 8 activities. The four activities at the bottom of the table represents a break down of Model refinements, cf. the timeline table.

Model development	Allocation	Time spend	
Documentation		15.0	3.9
User guide		5.0	0.0
Publish		3.0	0.0
Promote		5.0	0.0
Data and calibration		13.0	0.0
Energy markets		5.0	0.0
Model structure		2.5	0.7
Abatement model		8.0	0.0
		56.5	**4.6**

At present the total budget of 500.000 DK is allocated at an estimated hourly cost of 4700 kr. per day, 7,4 work hours per day, and with an overhead of 5 %. The hourly cost only covers direct labour costs, and it depend on the specific employees carrying out the activities. We don't have a specific target for the overhead rate. 5 % is however to be considered low compared to the level of indirect labour costs and fixed costs of our institution.

Appendix A: Summary of Short-Term and Long-Term Goals with Support Needs (AI-generated)

1. Finland

Short-Term Goals (2026):

- **Complete National Dataset: Finalize the first version of the FinReform model dataset, derived from 2019 data, by Q1 2026.**
- **Initial Calibration: Start model testing and initial calibration as soon as the dataset is ready.**
- **Pilot Analyses: Produce relevant pilot analyses focusing on environmental taxation and transport fuels within the first year.**

Long-Term Goals:

- **Long ambitions include establishing a permanent national modelling consortium, potentially inspired by DREAM.**
- **Integration into Policy Processes: Integrate modelling tools into regular fiscal planning, climate strategies, and impact assessments.**

Need for Support from DREAM:

- **Technical Guidance: Assistance in understanding data structure and aggregation decisions to ensure they align with desired analyses.**
- **Calibration Assistance: Support is required after the initial dataset is ready to adapt the model to Finnish specifics and enhance parameter estimation.**
- **Model Interpretation: Guidance on interpreting model dynamics during pilot analyses to ensure clarity and effectiveness in policy-making discussions.**

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2. Austria

Short-Term Goals (2026):

- **Design a Data Pipeline: Successfully design and implement a fully scripted, reproducible data pipeline that aligns with the GreenREFORM framework.**
- **Establish Data Structures: Secure clarity in data structures and aggregation choices to facilitate immediate updates in model use.**

Long-Term Goals:

- **Robust Modelling Framework: Create a technically resilient and long-term updateable modelling framework that can support a wide range of policy analyses.**

- **Institutional Integration: Embed macroeconomic climate modelling into routine budgeting and fiscal policy processes as part of the Green Budgeting framework.**

Need for Support from DREAM:

- **Process Design: Technical support for establishing a comprehensive data pipeline and ensuring it meets all future analytical needs.**
- **Documentation Standards: Assistance in developing standards for documentation and transparency that aids in model calibration and usability.**
- **Training Workshops: Facilitate workshops aimed at enhancing understanding of modelling capabilities and methodologies among stakeholders.**

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3. Belgium

Short-Term Goals (2026):

- **Operationalize the Model: Ensure the GreenREFORM model can run on Belgian data and produce a stable baseline for analyses aimed at specific policy applications.**
- **Develop Technology Catalogues: Populate the abatement module with credible technology data relevant to Belgian contexts.**

Long-Term Goals:

- **Dynamic CGE Model: Establish a flexible, modular CGE model that can address a wide range of climate and environmental policy questions effectively.**
- **Cross-Departmental Use: Promote broader usage of the model across departments within the government for enhanced decision-making.**

Need for Support from DREAM:

- **Guidance on Calibration: Support in calibrating the GreenREFORM model on the Belgian dataset and ensuring it integrates effectively with existing models.**
- **Sharing Best Practices: Facilitate exchanges on successful methodologies for addressing complex modelling challenges related to climate and energy policy.**
- **Error Resolution Support: Technical assistance for resolving model-specific issues during the calibration phase.**

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4. Italy

Short-Term Goals (2026):

- **Integrate Data Sources: Consolidate and ensure internal consistency across the GreenREFORM-Italy dataset, including national accounts and energy balances.**
- **Initial Policy Shock Testing: Conduct exploratory tests on policy simulations (e.g., carbon pricing) by Q4 2026 to observe model behavior and dynamics.**

Long-Term Goals:

- **Embed as Core Analytical Tool: Institutionalize GreenREFORM-Italy within the fiscal planning and climate strategies of the Department of Finance.**
- **Strengthen Systematic Use: Foster systematic use of model outputs for macroeconomic and fiscal assessments, improving coherence in analyses.**

Need for Support from DREAM:

- **Methodological Guidance: Specific guidance on data aggregation methods and their implications on model calibration.**
- **Support for Colleague Exchange: Opportunities for face-to-face discussions and exchanges with other EU countries that have successfully implemented similar models.**
- **Knowledge Sharing: Exchange on calibration parameters and behavioral assumptions to enhance the robustness of the modelling framework.**

Conclusion

The success of the GreenREFORM project in each country hinges on achieving both short- and long-term goals centered around effective data development, modeling capabilities, and integration into policy frameworks. Comprehensive support from DREAM is crucial, providing the necessary expertise, technical assistance, and collaborative opportunities to ensure ongoing progress and refinement of each country's modelling initiatives. This collaborative approach aims not only to prepare for immediate analyses but also to secure sustainable analytical frameworks for the future.

Key Challenges in Each Country

1. Finland

- **Data Integration**: Finland faces challenges in building a cohesive national dataset that can efficiently integrate various sources of data while ensuring consistency and relevance.
- **Capacity Building**: There is a need for enhanced capacity within public administration to effectively utilize the advanced modeling tools, especially in areas like climate macroeconomic modeling.

- **Time Constraints**: The timeline for producing pilot analyses and model calibrations is tight, especially as the country aims to meet national policy deadlines before upcoming parliamentary elections.
- **Technical Support Accessibility**: Dependence on external support, particularly with understanding the structure of data and aggregation choices, can impede progress if not managed appropriately.

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2. Austria

- **Sustainable Data Management**: The need for a programmable and sustainable data pipeline is critical, as prior experiences indicated that ad-hoc data preparation is not viable for long-term modeling.
- **Resource Constraints**: Limited internal resources and personnel dedicated to modeling within the Ministry of Finance can hinder effective model calibration and operations.
- **Complex Integration Processes**: Integrating different datasets and ensuring they align with the GreenREFORM framework presents technical challenges that require careful planning and execution.
- **Stakeholder Engagement**: Gaining broader internal engagement across departments within the Ministry of Finance remains a challenge, as the model's utility spans multiple policy areas.

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3. Belgium

- **Model Calibration**: Calibrating the GreenREFORM model to work effectively with Belgian data poses significant challenges due to the dynamic nature of the dataset and its integration with the TIMES model.
- **Data Fragmentation**: Belgium has a rich but fragmented statistical landscape, leading to challenges in reconciling diverse datasets into a coherent model framework.
- **Technical Barriers**: There are intrinsic complexities in the model that necessitate ongoing support and guidance for smooth operation, particularly around error resolution and integration processes.
- **Resource Dependency**: The reliance on external partners, like VITO and DREAM, for complex modeling challenges could slow down progress if responsiveness is not consistent.

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4. Italy

- **Data Harmonization**: The key challenge for Italy lies in reconciling multiple high-resolution data sources into a unified dataset suitable for the dynamic CGE framework, which requires careful synthesis choices.
- **Integration of Models**: Strengthening the link between the CGE model and sector-specific energy models, such as TIMES-Italy, is crucial and presents a technical complexity that must be adequately addressed.
- **Policy Relevance vs. Detail**: Balancing the sectoral detail needed for policy analysis against the model's tractability and robustness is a continual challenge that requires thoughtful trade-offs.
- **Maintaining Consistency**: Ensuring the model remains consistent with national accounting and energy representation while answering policy-relevant questions adds to the intricacy of the modeling task.

Conclusion

Each country faces unique challenges in their data development and implementation plans for the GreenREFORM project. Addressing these issues will require targeted strategies, enhanced collaboration with DREAM, and the establishment of sustainable processes that balance technical requirements with policy relevance.

Appendix B: Samarbejdsaftale mellem KU og DREAM for ph.d.-projektet "Macroeconomic Modelling and the Green Transition"

UNIVERSITY OF COPENHAGEN
DEPARTMENT OF ECONOMICS

Samarbejdsaftale mellem KU og DREAM for ph.d.-projektet "Macroeconomic Modelling and the Green Transition"

28 JANUARY 2026

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Dette dokument indeholder en samarbejdsaftale mellem Danish Research Institute for Economic Analysis and Modelling (herfra DREAM) og Økonomisk Institut, Københavns Universitet (herfra KU) vedrørende ph.d.-projektet "Macroeconomic Modelling and the Green Transition".

Formålet med ph.d.-projektet er at udvikle generiske makroøkonomiske værktøjer, som kan anvendes på både industrialiserede lande og udviklingslande. Ph.d.-projektet har tre hoveddele:

1. Den første hoveddel består i at bygge en grøn dynamisk generel ligevægtsmodel for Vietnam inspireret af GreenREFORM modellen samt demonstrere, hvordan modellen kan bruges til politikudvikling. Denne første del af projektet bygger på modelrammen GreenREFORM EU, som DREAM har udviklet til brug af EU-lande.

2. Den anden hoveddel består i at videreudvikle delelementer af den overordnede modelramme GreenREFORM EU.

3. Endelig indeholder projektet et mere selvstændigt projekt inden for makroøkonomisk modellering og grøn omstilling. Den ph.d.-studerende vil være hovedansvarlig for at udvikle denne del af ph.d.-projektet.

Alle dele af ph.d.-projektet kan involvere samarbejde mellem KU og DREAM. KU forventes at tage initiativet ift. første hoveddel, DREAM forventes at tage initiativet ift. anden hoveddel, og den ph.d.-studerende forventes at tage initiativet på tredje hoveddel. Det forventes, at hver hoveddel har det samme tidsforbrug, som vil være på knap et år givet ph.d.-projektets længde (3 år) samt den ph.d.-studerendes yderligere forpligtigelser.

I den anden hoveddel af projektet, vil den ph.d.-studerende bistå DREAM i at videreudvikle GreenREFORM EU modelrammen. Dette delprojekt har både et videnskabeligt og et mere praktisk sigte. I forhold til det praktiske skal dette delprojekt munde ud i, at GreenREFORM EU modellen kan bruges til at analysere flere typer politiske tiltag og/eller bruges på flere lande. Det videnskabelige sigte består i, at delprojektet munder ud i en videnskabelig artikel, hvor der sigtes efter udgivelse i et anerkendt internationalt økonomisk tidsskrift.

KU finansierer de første to år af ph.d.-projektet, mens DREAM finansierer det tredje år. DREAM finansierer det tredje år enten direkte via en overførsel til KU eller indirekte i form af en underleverandøraftale med KU. Det forventes, at DREAMs udgift til ph.d.-projektet bliver ca. 540.000 kr.

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